

CHAPTER – SEVEN

WELFARE ECONOMICS

A consumer is at equilibrium when he gets maximum satisfaction. A consumer gets maximum satisfaction when he equates the marginal utility derived from his consumption with the price of the commodity he consumed, or when the marginal utilities derived from two commodities are equal or the marginal rate of substitution between two goods is equal to their price ratio. Generally economists hold the view that when one's satisfaction increases, he is at a better welfare position than before. But the next question arises, as to the level of production or distribution or consumption, at which the whole society's welfare is better-off. We have discussed in the exchange chapter that perfectly competitive market situation is good for the society. The society is in a better position if the output is maximum. But, to certain economists, if a policy brings more benefit to the society, then the society will be better-off in terms of welfare. It is only in the last quarter of the nineteenth century, economists began to distinguish between economic theory and policy. It is also said that the happiness is the maximum when we are able to enjoy maximum satisfaction from any activity, including consumption. Welfare is more in a state of affair in comparison with another state of affair and vice versa. The state of affair, which yields maximum welfare, is accepted and implemented by the authority for the best advantage of the society.

Welfare economics is, thus, concerned with the evaluation of alternative economic situations. It is good for the society trying to reach the maximum well-being it can reach with the given factor endowments and the state of technology. For example, if there are two alternatives A & B to a particular factor endowment and if B can yield more well-being than A, then it will be good for the society to accept the B alternative and adopt it. To evaluate alternative economic situations, a number of criteria have been developed by various economists like Pigou, Pareto, Kaldor, Hicks, Scitovsky, Bergson, Samuelson, Arrow and Rawls.

examples of **external diseconomies**. But, the presence of public parks, street lights and public roads are good examples of **external economies**. Externalities, thus, exist when the means that influence the gain or cost are not marketable. These means are called **untraded interdependencies**, which may be unidirectional or reciprocal. Hence, we can infer that externalities are market imperfections, where the market offers no price for the benefit or cost or for the service or disservice. As there is market failure, it leads to misallocation of resources and thereby a situation where social well-being is not maximised.

Externalities or the divergence between private and social cost and return are due to external economies in production and in consumption, external diseconomies in production and in consumption and the presence of public goods. Production and consumption of certain goods may result in an increase in welfare in comparison with the production cost or private cost. For example, if neighbours are allowed to see programmes in a TV purchased by an individual household then the total gain will be more than the private cost. This is similar to production of lead-free petrol, which increases the overall benefits to the society. But, production and consumption of cigarettes cause a serious health problem and so the social cost is more than the individual cost or production cost.

An improvement in social welfare takes place when the goods, which provide more social gains than private costs are produced more, at the expense of goods, which are having high diseconomies, and the poor are allowed to enjoy more share of the income than before. For example, construction of roads and irrigation facilities in rural areas at the expense of production of unwanted goods or goods with high disexternalities is an improvement in the welfare of the rural mass. The next question to be answered is how it can be achieved. There are many ways to achieve this. Taxes may be imposed on goods, which have high diseconomies, and the tax money can be used for other social welfare programmes. High tax will certainly reduce the attraction of investment in these types of goods. Thereby, the output of such types of goods can be reduced. Subsidy can also be provided to the producers of goods, which give more social benefits than private costs. It will favour an increase in the output of such commodities. Public goods can be supplied through public authority and people may be allowed to share the costs. Internalising of externalities is another source to attain market perfection. **Internalisation** means inclusion of social cost in the production cost/private cost. When ownership rights are well-defined, there is good chance for attaining market perfection (**Coase Theorem**).

7.1.4 Pigou's Ideal Output and Social Welfare

Ideal output refers to that level of output at which the social welfare is maximum. Generally, the ideal output is reached automatically under perfect competition. A firm is called optimum firm when its output is at the minimum average cost. When the product transformation curve is introduced, the iso-revenue curve is tangent to the product transformation curve. All these have been discussed under the production chapter. But, now, it is necessary to introduce the indifference curve and price line for finding out the ideal output for the whole economy. As it is reinterpreted by Baumol, the ideal situation of maximum welfare is available, only when the slope of the product transformation curve is equal to the price line and it, in turn, is tangent to the indifference curve. The ideal output

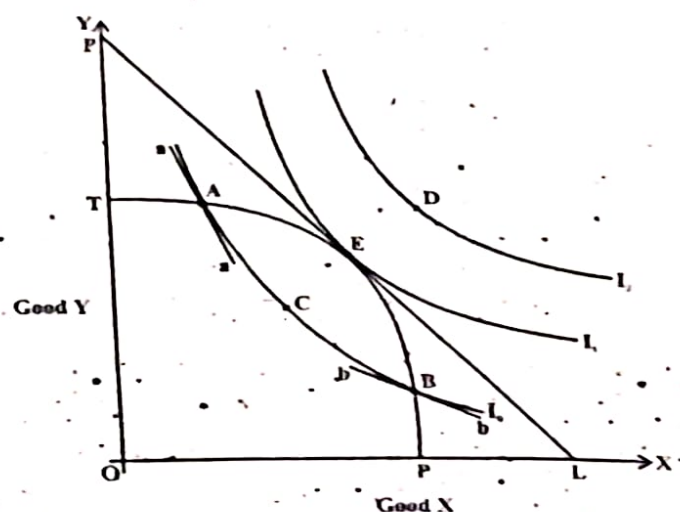


Figure 7.1

7.1 Pigovian Welfare Economics

The gist of welfare economics is contained in the standard work of A.C. Pigou, – ‘The Economics of Welfare’. A.C. Pigou is considered as the father of welfare economics. His views were first expressed in his book ‘Wealth and Welfare’ (1912). But, his entire views were explained in his book ‘The Economics of Welfare’ (1932) and it is known as the Pigovian neo-classical welfare economics. The Pigovian Welfare economics can be discussed in the following lines.

7.1.1 Meaning of Welfare

Pigou’s approach to welfare economics is very pragmatic. To him, welfare depends upon one’s state of mind regarding the satisfaction or utilities received from his consumption. Individuals’ welfare is determined on the basis of the extent to which an individual’s wants are satisfied. He limits his explanation only to that part of social welfare that can be brought directly or indirectly into relationship with the measuring rod of money. This is called economic welfare, i.e., welfare or satisfaction derived by an individual from the use of exchangeable goods and services. It means that he talks only about economic welfare. And so he says economic welfare is a part of total or general welfare and is a function of goods and their relative prices. Social welfare is calculated simply by summing up all individuals’ welfare. But, he has failed to take into account other forms of welfare, which depends on many other elements like the working condition, security, housing status and human relationship.

7.1.2 Welfare Conditions

Pigou adopts a dual criteria for judging the changes happening in the social welfare. They are called dual conditions for maximum social welfare. They are as follows:

- (i) **Maximisation of national income:** According to Pigou, if national income increases without a corresponding change in the supply of factors, then there will be an increase in the social welfare.
- (ii) **Transfer of national income from the rich to the poor:** Welfare is maximum when the share of the poor increases without affecting adversely the total income or the given national income.

The second welfare criterion is based on the dual, Pigovian postulates of ‘equal capacity for satisfaction’ and diminishing marginal utility of income. In short, he assumes that people receive equal satisfaction from equal real income and comparatively rich people have less capacity for enjoyment when compared to the poor. Therefore, every transfer in favour of the poor will add welfare to the existing well-being.

7.1.3 Externalities

Pigou points out that maximum welfare is not achievable because of market failure. The main cause for market failure is the presence of externalities. Externalities are said to be present if economic or other interactions create social gains or costs beyond those taken into account, by those involved in the interaction. We can say that the divergence between private and social costs and gains are externalities. Externalities may be external economies or external diseconomies. They are also called spill over or neighbourhood effects. The standard example of external diseconomies is environmental damage from production activities, causing pollution. Smoke emanating from brick kilns or untreated industrial effluents coming out of an industrial unit or domestic sector or hospital wastes dumped in the open area are good

examples of **external diseconomies**. But, the presence of public parks, street lights and public roads are good examples of **external economies**. Externalities, thus, exist when the means that influence the gain or cost are not marketable. These means are called **untraded interdependencies**, which may be unidirectional or reciprocal. Hence, we can infer that externalities are market imperfections, where the market offers no price for the benefit or cost or for the service or disservice. As there is market failure, it leads to misallocation of resources and thereby a situation where social well-being is not maximised.

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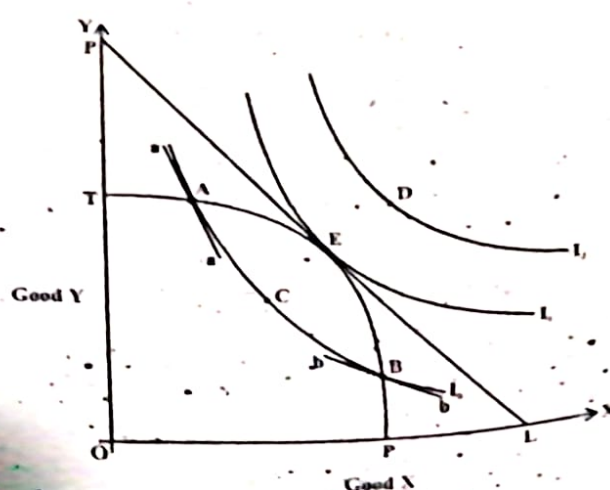


Figure 7.1

condition is attained only when all these three lines/curves are tangent. It is noted that the ideal output is that level of output at which no reallocation of the economy's resources among various uses would make the society better off than before. This situation is explained with the help of a figure, figure 7.1.

In figure 7.1, I_0 , I_1 and I_2 represent community indifference curves. PL is the price line and TP is the product transformation line. The ideal output is reached at the point E, where IC_1 is tangent to both PL (price line) and TP (product transformation line) lines. In other words, at point E, $MRPT_{xy}$

$= \frac{P_x}{P_y} = MRS_{xy}$. It is so only when there are no externalities. If there are externalities, then it is not

possible to reach the point E. It is possible to reach either 'A' or 'B' depending upon, which commodity comes under externalities. If 'X' is under external economies, then A is the equilibrium point. It is to the left of E and indifference curve is lower or to the left of I_1 . It means lower welfare. If Y is under external economies, then B is the equilibrium point. However, it is noted that in the first situation, less quantity of X is produced whereas in the second situation, less quantity of Y is produced than the ideal level.

7.2 Pareto's Optimal Condition or the Pareto Optimality Criterion

The Italian economist Vilfredo Pareto, a disciple of Leon Walras, has given the most innocuous value system, in which economic efficiency can be objectively measured. This criterion is otherwise called Pareto unanimity criterion or Pareto unanimity rule or Pareto criterion or Pareto optimality. According to this criterion, any change that makes at least one person better-off and no one worse-off is an improvement in social welfare. But, a change that makes no one better-off and at least one worse-off is a decrease in social welfare. In other words, when reallocation of resources is made, if there is a possibility for an individual to increase his welfare without affecting anyone, then the situation after reallocation is better than the situation before reallocation. It may be explained in a different way also. By Pareto optimality we mean a situation in which no one can be made better-off without making some one worse-off. It is the optimal situation and any shift or movement from this situation is not optimal. This situation is said to be Pareto-optimal or Pareto-efficient condition.

To attain this Pareto-efficient situation in an economy, three marginal conditions have to be satisfied. They are as follows:

- (i) Efficiency of distribution of commodities among consumers (consumption efficiency),
- (ii) Efficiency in the allocation of factors among firms (production efficiency), and
- (iii) Efficiency in allocation of factors among commodities (efficiency in the product-mix or composition of output).

7.2.1 Consumption Efficiency or Efficiency of Distribution of Commodities among Consumers

Consumption efficiency requires that it is impossible by a redistribution of a given set of goods to increase the utility of one individual without reducing utility of the other. It is easy to examine a community of two people, A & B who consume two commodities, water (X) and a composite of all other goods (Y). In response to a drought in their economy in mid 2010s, the authority announces water rationing. Initial consumption of 'A' and 'B' was different. Now they have been allotted equal amounts of water and it is assured that there is no sale of water between

individuals to avoid profiteering. This situation is not optimal as some reallocation or changes in policy improves one of the two persons without affecting the other. The details are given in the following figure 7.2.

The box figure 7.2 is the Edgeworth's box diagram, used for explaining exchange between two individuals. A's initial consumption, (before the introduction of water rationing), of all other goods is Oa and B's is $O'b$. Now, due to the drought in their economy, water rationing is introduced and no transfer of water or sale of water is allowed in the system. Point 'h', in the figure, indicates the resulting allocation.

The point is half way between point a and b and so it is easy to infer that each one is allocated equal amount of water. It is very important to note that the point 'h', is not on the contract curve. Contract curve refers to the curve joining all the tangency points between two indifference curves. It is the locus of Pareto-optimal or efficient distribution of goods between consumers. Moving up or right means that A is better-off and moving down means B is better-off. Any point on the contract curve is better than any point off the curve. Only on the contract curve, the condition for equilibrium, i.e.,

$MRS_{xy}^A = MRS_{xy}^B$ is satisfied. The point 'h' is not on the contract curve. It means that there is a possibility for improvement in welfare. Allowing trade would enable one or both of the two individuals to be better-off without any one being made worse-off. For example, if B is allowed to sell 'mn' litres of water to A, in return for 'hm' units of all other goods, they, both would be happier, i.e., both A and B move from their original indifference curves, I_{A0} and I_{B0} to a higher position at point n where I_{A1} is tangent to I_{B1} . It is very clear that from the explanation that allowing people to sell water is beneficial to both persons – the seller and the buyer. If they are allowed to reach the point I, then B is much benefited at the cost of A and so it is not acceptable to A. Hence, movement in between point p and q are better welfare points than before.

7.2.2 Production Efficiency or Efficiency of Allocation of Factors among Firms

The consumption efficiency, discussed above, assumes that the commodities are fixed in supply. Here also, it is assumed that the quantities of the inputs devoted to the production of each commodity are fixed. Production efficiency requires that production reaches a level where it is not possible by a reallocation of factors to produce more, than changing the combination of total output. By reallocation of inputs, the combination of total output can be changed but not the total output. The argument is closely analogous to the one used for the analysis of consumption efficiency. In consumption efficiency, it is necessary to equate the marginal rate of substitution between X and Y goods of A with that of B. In the same way for attaining production efficiency, it is necessary to attain a situation where the marginal rate of technical substitution between two factors, L and K, of X

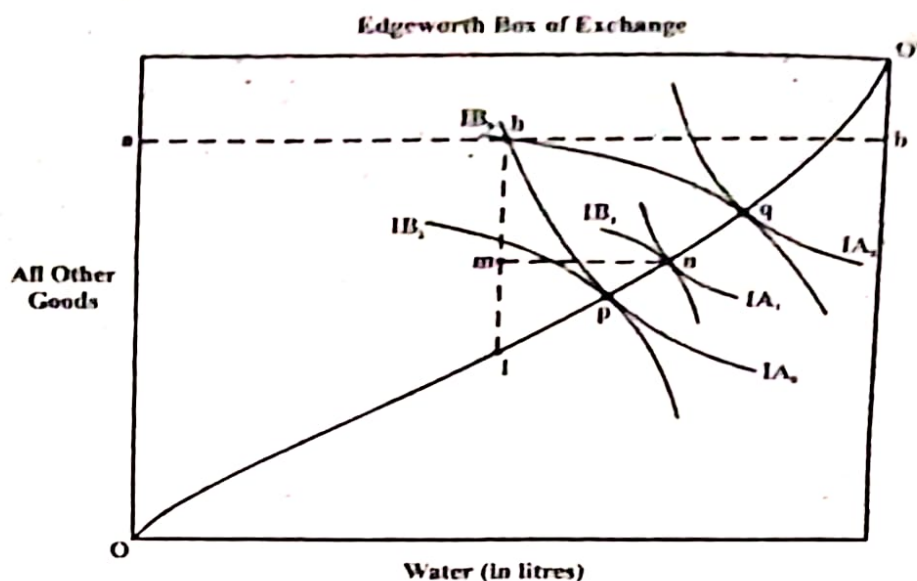


Figure 7.2

commodity must be equal to that of Y commodity. To explain this point, we use the same Edgeworth box figure, two inputs, K and L and two commodities X and Y. The production efficiency can easily be explained with the help of figure 7.3.

As discussed in the previous section, only points on the contract curve are Pareto-efficient. Hence, it is clear from figure 7.3, that point NE is not optimal or inefficient. Point NE is inefficient because a reallocation of the given K and L between X and Y goods, any point from a to c inclusive, leads to an increase in at least one commodity without a reduction in the other. For example, a movement from NE to point 'a' means an increase in the production of Y goods. The firm producing Y commodity moves from I_{Y1} to I_{Y2} , a higher isoquant. The point NE indicates no exchange position or position before reallocation. But, if they move from NE to 'c' then firm producing X commodity increases its output without affecting the output of Y. On the other hand, if they are able to move to point b, both firms are benefited. However, it is noted that any point on the contract curve satisfies the optimality condition.

i.e., $MRTS_{LK}^X = MRTS_{LK}^Y$. It means that the marginal rate of technical substitution between labour and capital of X firm must be equal to the marginal rate of technical substitution between labour and capital of Y firm. But, it should be noted that before the reallocation, NE is the point i.e., X is in I_{X0} and Y is in I_{Y1} and so they are not ready to move down from the point NE but ready to move up from this point. Hence, any point between a and c inclusive is acceptable to both the firms X and Y, depending on the bargaining power of individual firms.

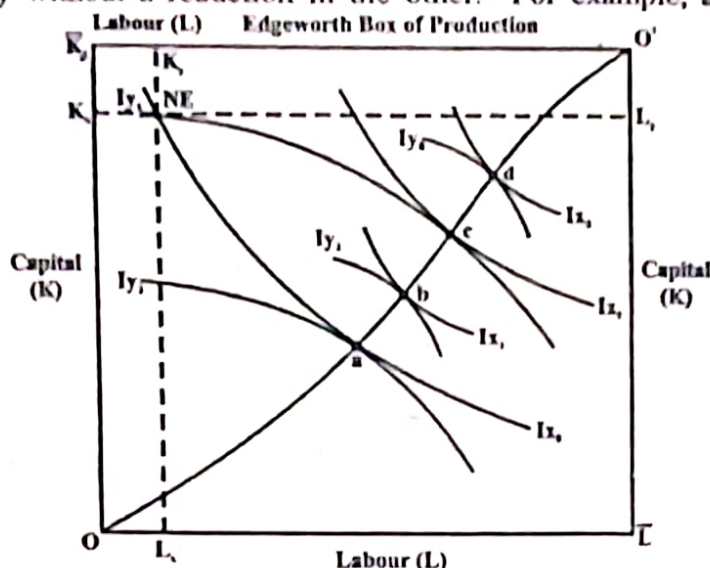


Figure 7.3

7.2.3 Combining Consumption Efficiency and Production Efficiency or Efficiency in Product-mix or Efficiency in the Composition of Output

The third possible way to maximise welfare is by changing the product-mix. When it is possible to maximise social welfare by merely changing the product-mix, then it is conducive to the whole society. The third marginal condition to reach maximum welfare is that the slope of the production possibility curve must be equal to the slope of the exchange curve. The slope of the production possibility curve (PPC) is called the marginal rate of product transformation curve between two goods X and Y ($MRPT_{xy}$). The slope of the exchange curve is called the marginal rate of substitution between two goods X and Y. Symbolically,

$$MRPT_{xy} = MRS_{xy}^A = MRS_{xy}^B$$

It is very clear from the above formula that the marginal condition for a Pareto-optimal composition of output requires that the MRPT between any two goods must be equal to the MRS

between the same goods. Like the other two marginal conditions, this condition is also illustrated with the help of a figure 7.4.

As it is discussed in figure 1.1, production at any point inside the PPC is inefficient and only on the PPC it is efficient. When we combine production and consumption, Pareto-optimal criterion would refer to a situation where it is not possible to increase the welfare of one individual without reducing the welfare of any other individual, by producing more of one good and less of another good by reallocating resources. In figure 7.4, PT curve indicates the economy's PPC and I curves represent the welfare of the economy by indifference curves, I_0 , I_1 & I_2 . The point E in the figure 7.4 represents Pareto-optimal state from the view point of both production and consumption. But, the other points A, B and C are not optimal from the point of view of the economy as a whole, because as it is said, the point C is inside the PT curve and points A and B are related to a low indifference curve, where the slopes are not equal. Further, it is obvious that movement from A to E or B to E increases the welfare of the economy as a whole. At the point E the marginal condition that $MRPT_{xy} = MRS_{xy}^A = MRS_{xy}^B$ is satisfied.

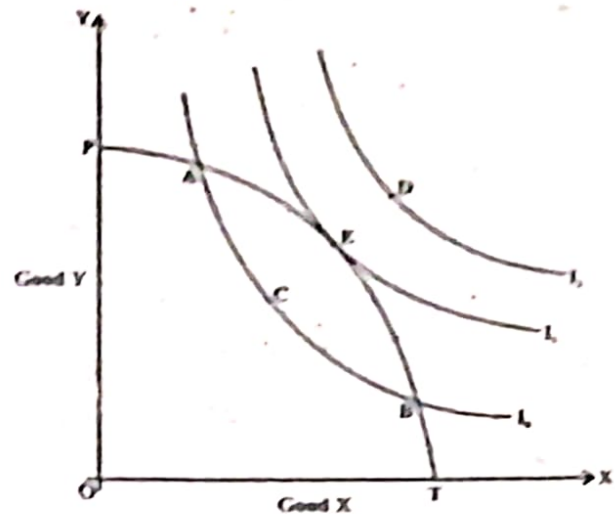


Figure 7.4

Though point D is at a still higher social indifference curve, I_2 , that point is not attainable for the given resources. The maximum reachable points are indicated by the product transformation curve PT. Hence, E represents Pareto-optimality from the point of view of the economy as a whole. This condition must be fulfilled for each commodity both in production and in consumption. In the problem of distribution of goods or exchange, the condition is that MRS_{xy} must be equal for every consumer and in the problem of allocation of inputs in production, the condition is that $MRTS_{LK}$ must be equalised in the production of each commodity. But for the production of an optimal product-mix, MRS in consumption must be equal to $MRTS$ in production.

7.3 Value Judgement

Value judgement literally means subjective estimation. Simply, we can say value judgement is some ethical belief and judging a situation or a state of affairs on the basis of the ethical belief. All ethical statements recommending, influencing or persuading are simply called value statements or judgements. They, generally, describe the facts in an emotive way and tend to influence people by simply altering their attitudes. As value judgements describe facts, but are prescriptive in nature, they are, hence, qualitative statements. To attain the optimum position, every individual acts as the best judge of his welfare. To reduce or avoid smoking, recently, the government of India has passed an Anti-smoking Act. But, from the point of view of smokers, their satisfaction is reduced by this Act. But, ethically and from the government's and the society's point of view it is a very good step. From the point of society, the government likes to have a conducive environment in places where people gather. Terms like poverty, eradication and inequality reduction are good examples of value judgements. They increase the economic welfare or welfare of the whole society. Hence, it can be

concluded that welfare propositions involve value judgements. In other words, welfare economics involves value judgements and thereby interpersonal comparisons.

Economists differ in their ideas over this issue. The important questions that arise are, should economists make use of value judgements in welfare economics? and whether it is possible to measure welfare without doing interpersonal comparisons?

It is generally accepted that as welfare is an ethical term, all welfare propositions are ethical and involve value judgements. For example, terms such as social, community, national, satisfaction and utility are highly emotive and so economics is not free from value judgements. As welfare economics is concerned with policy measures, it involves ethical terminology such as social welfare, social benefit and social advantage.

The classical economists viewed welfare economics without using any value judgements. For example, according to classical economists, welfare of a nation varies directly with its level of production. But, the neo-classical economists like Marshall are concerned with the cardinal measurement of utility and they think that interpersonal comparison of utility is inevitable. Pigovian welfare economics is also based on this principle. But, Robbins in 1932 held the view that there is no necessity for interpersonal comparisons and welfare can be maximised by fulfilling Pareto-optimality conditions. It means that there is no necessity for making interpersonal comparisons to maximise welfare. On the basis of this view, Kaldor, Hicks and Scitovsky formulated the compensation principle.

Bergson, on the other hand, points out that the compensation principle itself has involved some value judgments. According to him and Samuelson and Arrow, there is no welfare economics and no meaningful welfare proposition can be made without introducing value judgements. According to them the Pareto's optimal criteria are not value-free. Boulding's view has put an end to this controversy. According to Boulding, the statement of Pareto that no body should ever be made worse-off itself involves interpersonal comparisons and so the Pareto-optimality itself is a special case of interpersonal comparison for maximising social welfare function. Hence, it is not possible to avoid interpersonal comparisons in welfare economics.

7.4 Social Welfare Function or The Bergson Criterion

A social welfare function (SWF) is simply a rule for measuring aggregate welfare from individuals' welfare. So far we have discussed that a change is considered to be good when every one benefits without hurting any one. But, it is impossible to evaluate when a change brings benefits to some individuals at the cost of some others. In other words, the welfare criteria so far discussed show that when a change in the economic policy benefits some persons and hurts some other persons, it is impossible to evaluate it without making some value judgement about the deservingness of the different individuals or groups. Abram Bergson has suggested an explicit set of value judgements in the form of a social welfare function to measure the welfare of different individuals and to find out the net results. A social welfare function is analogous to the individual consumer's utility function. It provides a ranking of alternative states, in which different individuals enjoy different utility levels. It tries to arrive at a social ordering of feasible social states from individuals' ranking, in terms of their desirability.

The concept of social welfare function was first introduced by Bergson and later on scientifically developed by Samuelson, Tinter and Arrow. Bergson takes an economy consisting of two individuals and explains the social welfare function by a set of social indifference contours. The

social indifference contours are not smooth curves like the ordinary indifference curves but in meaning they are the same. A curve is a locus of points representing constant or same social utility or welfare. Further, like the ordinary indifference curve, the social indifference contour to the right side indicates higher welfare than the leftward social indifference contour. Hence, a change, which could move the society from a point on the lower indifference contour to a point on the higher indifference contour increases the social welfare and vice versa. The details are illustrated in the figure 7.5.

As given in the figure, the social indifference curves are not smooth. They have some ups and downs depending upon the relationship between welfare and utilities of individuals A and B. To the whole society, a movement from 'a' to 'b', has no net benefit. In other words, the movement from a to b or b to a leaves the level of social welfare unaltered. But, a movement from a to c or a to d or b to c or c to d is an increase in the social welfare. But, however it is very difficult to compare different individuals and rank them.

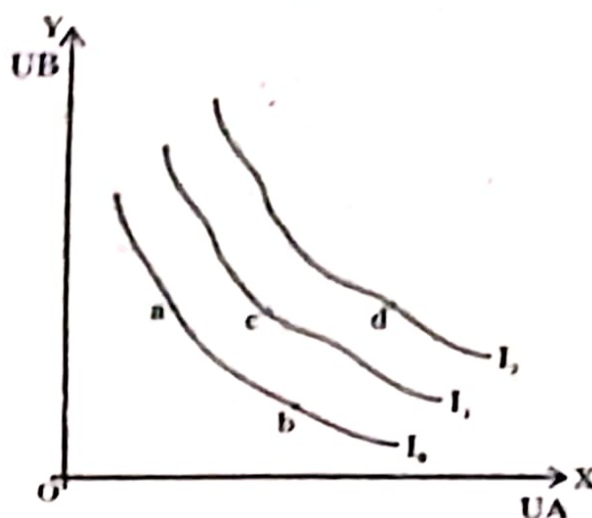


Figure 7.5

7.4.1 Maximisation of Social Welfare or Determination of the Point of Bliss

The point at which the social welfare is maximum is called the **bliss point**. In this section, a discussion is carried out to examine the conditions of social welfare maximisation. Here, to simplify the explanation, a two-factor, two-commodity, two-consumer model is used. Thus, there are two factors, Labour, L and Capital, K and their quantities are given. There are only two goods X & Y to produce (each firm produces only one commodity) and there are only two consumers A and B. The motive is utility maximisation for consumers and profit maximisation for producers. There is no economies of scale and the utilities of consumers are independent.

To find out the point of bliss we have to use two curves; viz., (i) social welfare function and (ii) grand utility possibility frontier. The social welfare functions are used to find the situation where the social welfare is maximum. The grand utility possibility frontier shows the maximum utility attainable by B, given the utility enjoyed by A from any given product-mix. Each point on the PPC gives rise to a utility possibility frontier. The envelope of the various utility possibility frontiers is the grand utility possibility frontier of the economy. Maximum social welfare is attained at a point where the grand utility possibility frontier curve is tangent to the highest possible social welfare function or social indifference contour. In other words, social welfare is maximum at the point of tangency of the envelope utility possibility frontier with the highest possible social indifference contour. The tangency point is called the

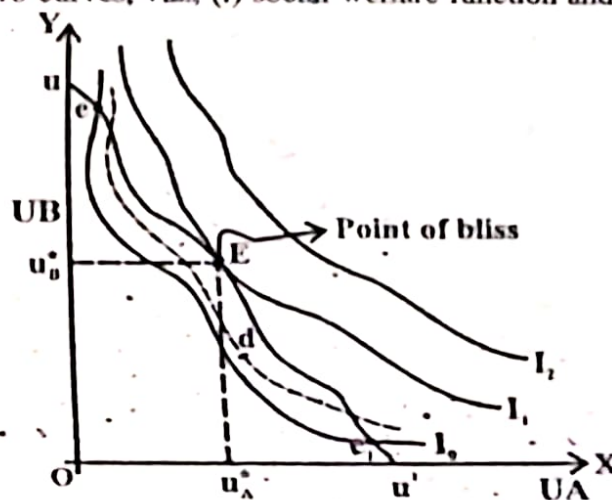


Figure 7.6

avoids making inter-personal comparisons of utility. Most economic changes are of the type Kaldor-Hicks visualised and so Kaldor-Hicks criterion is practically valid.

First we shall explain the Kaldor-Hicks criterion with a numerical example, then move to graphical illustration. Suppose, there is a plan to widen the road in a village, the programme will enable all villagers to enjoy benefits like public transport. But, a few land owners have to lose some area of their land. The gainers are asked the question how much they are ready to pay to carry-out the project. At the same time the losers are also asked the question how much they are ready to pay for preventing the project. If they are ready to pay one lakh rupees and the villagers are ready to pay more than one lakh rupees, then the Kaldor-Hicks criterion is satisfied and so the project is justified. This criterion is illustrated with the help of a figure 7.7.

The figure 7.7 is drawn to explain the change with two individuals A and B and two bundles of commodities, X and Y. The two utility possibility curves AB and A_1B_1 represent the two bundles of goods. Starting our analysis from point a, A's move from a to b or c or between b and c is an improvement in social welfare, satisfying the Pareto optimality condition. Any point between c and b inclusive represents a change that benefits atleast one without hurting the other or benefits both the individuals. The same is the result if they move from a_1 to b_1 or c_1 or between b_1 and c_1 . But, any movement outside b and c or b_1 and c_1 or movement from a to a_1 or a_1 to a cannot be evaluated by the Paretian criterion because it improves A's welfare or B's welfare at the expense of B or A. It can be evaluated in terms of Kaldor-Hicks criterion. For example, if a change is planned and there is a move from a to a_1 , it is possible to compare welfare only with the help of Kaldor-Hicks criterion. It is beneficial to A at the expense of B and so the gain is more than the loss, and then it is an improvement in social welfare. It is very interesting to note that there is a possibility for moving to point d. Point ' a_1 ' is beneficial to 'A' only under the Kaldor-Hicks condition that ' a_1 ' (new bundle) must be above the point 'a' (initial bundle).

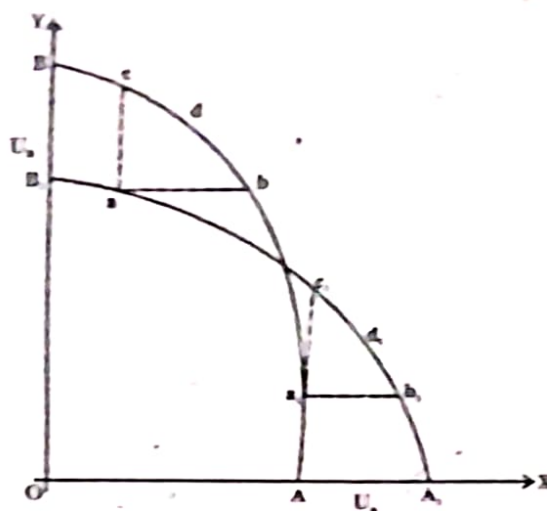


Figure 7.7

Thus, Kaldor-Hicks criterion satisfied what the Paretian criterion did not satisfy. But it is also not free from limitations. It failed to indicate a common standard value and income distribution is completely ignored. This criterion because of its difficulty in application loses its universal validity. Further, it talks about only potential compensation but not actual compensation.

7.5.2 Scitovsky Criterion

It was Tibor Scitovsky (1941) who pointed out the limitations of the Kaldor-Hicks criterion. He pointed out that the Kaldor-Hicks criterion leads to a contradiction. As given in figure 7.7, a movement from 'a' to ' a_1 ' is an improvement, satisfying the Kaldor-Hicks criterion. But, the counter move from ' a_1 ' to 'a' also passes the Kaldor-Hicks criterion and thereby shows an improvement in social welfare. In such a case, each situation would be better than the other. This contradiction is called Scitovsky paradox. Scitovsky has suggested another criterion called Scitovsky reversal test to overcome this contradiction. According to this criterion, a change is said to increase welfare if,

point of bliss. In the figure 7.6 given below, the grand utility possibility frontier is combined with the social welfare function represented by a number of social indifference contours.

The social welfare is maximum at the point E, which is called the point of bliss. The grand utility curve uu' is the locus of utility combinations which satisfy the marginal condition $MRPT = MRS$ for each commodity-mix. The slope of this utility frontier must be equal to the slope of the social indifference contour. The social indifference contour I_1 is tangent to the grand utility possibility frontier at the point E where the two slopes are equal.

Now, we shall discuss why Bergson criterion is superior to Pareto optimality. It is first indicated that Pareto optimality is a necessary condition but not a sufficient condition for welfare maximisation. All the points on the grand utility possibility frontier satisfy all the three conditions of Pareto-optimality. But, Bergson says only the point E yields maximum social welfare and not all the points on the utility frontier. For example, point e and e_1 are on the utility frontier and so these points must give maximum social welfare. But, it is not so because at point 'd', the social welfare is more than at point e and e_1 as point 'd' is on the higher indifference contour. Hence, Pareto optimality is a necessary condition but not a sufficient condition.

7.5 Compensation Principle or The Kaldor, Hicks and Scitovsky Compensation Criterion

The compensation criteria were originated to provide a basis for making welfare comparisons in situations where some gain and some others lose, when redistribution or reallocation occurs. When a change occurs in an economy and in the resultant situation some individuals gain and some others lose, it is difficult to find out, whether the benefit is maximised or not. To combat such a stalemate, a new set of welfare criteria have been developed by economists like Kaldor, Hicks and Scitovsky. All the three economists mentioned above were able to explain maximisation of social welfare, without making value judgements. Tibor Scitovsky's model is somewhat different from the model developed by Nicholas Kaldor and John Hicks. But, both the models are based on compensation principle and so all the models are called in the name of compensation principle.

7.5.1 The Kaldor-Hicks Criterion

It is very difficult to bring an economic change, which brings only favourable effects on individual welfare. Any policy change aiming at greater economic development and economic welfare has resulted in some disadvantages to atleast a few. If efficiency used by Pareto is the system of comparison of welfare, then there will be no economic policy changes. Hence, a model to measure the net welfare is developed by Kaldor and Hicks. The Kaldor-Hicks criterion says that a move from social state 'A' to another social state 'B' is considered as an improvement in social welfare, if the gainers from such a move have some extra gain, after compensating the loss incurred by losers. In other words, assume that an economic change is contemplated, which will benefit some and hurt others. Those persons who are benefited by the change are called gainers and persons hurt are losers. If the gainers are ready to pay more for carrying out the change than the losers, who are ready to pay for preventing the change, then the change will constitute an improvement in social welfare because the gainers could compensate the losers and still have some net gain. If the gainers can bribe the losers for accepting the change, Kaldor-Hicks criterion is met and so the change is justified. Here, it is assumed that the marginal utility of money remains constant.

Kaldor-Hicks criterion is, thus, applied to a situation, where an economic change benefits some people and hurts others. In such a situation, the Pareto criterion cannot be applied since Pareto

avoids making inter-personal comparisons of utility. Most economic changes are of the type Kaldor-Hicks visualised and so Kaldor-Hicks criterion is practically valid.

First we shall explain the Kaldor-Hicks criterion with a numerical example, then move to graphical illustration. Suppose, there is a plan to widen the road in a village, the programme will enable all villagers to enjoy benefits like public transport. But, a few land owners have to lose some area of their land. The gainers are asked the question how much they are ready to pay to carry-out the project. At the same time the losers are also asked the question how much they are ready to pay for preventing the project. If they are ready to pay one lakh rupees and the villagers are ready to pay more than one lakh rupees, then the Kaldor-Hicks criterion is satisfied and so the project is justified. This criterion is illustrated with the help of a figure 7.7.

The figure 7.7 is drawn to explain the change with two individuals A and B and two bundles of commodities, X and Y. The two utility possibility curves AB and A_1B_1 represent the two bundles of goods. Starting our analysis from point a, A's move from a to b or c or between b and c is an improvement in social welfare, satisfying the Pareto optimality condition. Any point between c and b inclusive represents a change that benefits atleast one without hurting the other or benefits both the individuals. The same is the result if they move from a_1 to b_1 or c_1 or between b_1 and c_1 . But, any movement outside b and c or b_1 and c_1 or movement from a to a_1 or a_1 to a cannot be evaluated by the Paretian criterion because it improves A's welfare or B's welfare at the expense of B or A. It can be evaluated in terms of Kaldor-Hicks criterion. For example, if a change is planned and there is a move from a to a_1 , it is possible to compare welfare only with the help of Kaldor-Hicks criterion. It is beneficial to A at the expense of B and so the gain is more than the loss, and then it is an improvement in social welfare. It is very interesting to note that there is a possibility for moving to point d. Point 'a₁' is beneficial to 'A' only under the Kaldor-Hicks condition that 'a₁' (new bundle) must be above the point 'a' (initial bundle).

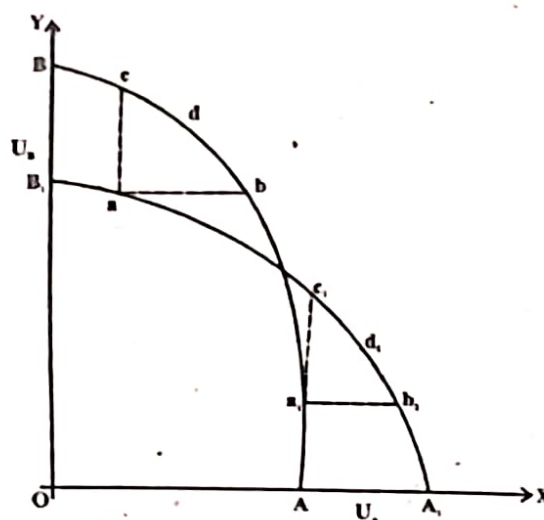


Figure 7.7

Thus, Kaldor-Hicks criterion satisfied what the Paretian criterion did not satisfy. But it is also not free from limitations. It failed to indicate a common standard value and income distribution is completely ignored. This criterion because of its difficulty in application loses its universal validity. Further, it talks about only potential compensation but not actual compensation.

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- (i) Kaldor-Hicks criterion is satisfied and
- (ii) If the losers are incapable of bribing the gainers for accepting the change.

Hence, a movement from point 'a' to point 'c' is an improvement because both A and B can move further to point d. At point 'd' both are better off than before. But it is not 'c' to a_1 .

Scitovsky, thus, solved the contradiction of Kaldor-Hicks criterion and took the welfare economics to a new height. However, the compensation criterion is concerned with only efficiency aspect of social welfare, at the neglect of equity. The question of equity is concerned with the relative preference of individuals in the whole economy. It also leads to a contradictory outcome and fails to provide any clear ordering of available economic alternatives. Another failure of this model is, it assumes constant marginal utility of money. But, it is a well known fact that the marginal utility of money is not constant and so redistribution resulting in inequity cannot be properly measured unless we know the exact marginal utility of money to different individuals.

7.6 Inability to Obtain Optimum Welfare

It is generally accepted that competitive market leads to an efficient allocation of resources at a given point of time. Every competitive equilibrium is Pareto-optimal. It is the first welfare theorem called Invisible Hand Theory or Harmony Thesis. The second welfare theorem states that a Pareto optimum allocation of resources can be realised if one uses a pricing mechanism in perfectly competitive markets. The essential of competitive equilibrium is that at a set of price ($P_1, P_2, \dots, P_n$) all markets are simultaneously cleared. Consumers' and Producers' equilibriums are mutually consistent with each other and in aggregate, all markets are cleared. Consumers maximise their satisfaction and producers maximise their profit and at equilibrium their maximising points coincide. The maximum happiness point (point of bliss) is reached where slopes of the isoquant, the indifference curve, the production possibility curve and the budget line are equalised. To say in simple terms, perfect competition can lead to a general equilibrium situation where the three marginal conditions of Pareto are satisfied. It is worthy to consider the following points here:

- (i) Profit maximisation is achieved when the following condition is satisfied, i.e., $MRTS_{LK} = \frac{w}{r}$
for individual firm and for all firms $MRTS_{LK}^X = MRTS_{LK}^Y = \frac{w}{r}$.
- (ii) Utility maximisation is achieved only if and when $MRS_{xy} = \frac{P_x}{P_y}$ for an individual and for the whole economy $MRS_{xy}^A = MRS_{xy}^B = \frac{P_x}{P_y}$.
- (iii) At the bliss point, the slope of the production possibility curve is equal to the slope of the indifference curve, i.e., $MRPT_{xy} = \frac{P_x}{P_y}$.
- (iv) A perfectly competitive market maximises its profit by equalising its marginal cost to the market price. Thus we have,

$$MRPT_{xy} = \frac{P_x}{P_y} = \frac{MC_x}{MC_y}$$

From the above analysis, it is inferred that a perfectly competitive system ensures the attainment of maximum welfare. But, if perfect competition is absent, then maximum social welfare will not be achieved. Further, it is also assumed that there is no externalities, no public goods, no increasing returns and no difficulty in dividing inputs. Though, for attaining maximum social welfare, the indifference curves and isoquants must be convex to the origin and the production possibility curve is concave to the origin and the demand functions must be continuous as these conditions are not very important, no separate discussion about these is not carried out here. But the conditions mentioned first are discussed one by one to analyse how the absence of these conditions prevents the society from reaching the maximum social welfare level.

7.6.1 Market Power or Absence of Perfect Competition

The first welfare theorem holds that all consumers and firms are only price takers and not price makers. If some individuals or firms influence the price or are price makers, then there will be no efficiency in resource allocation. It is because under perfect competition the price is equal to the marginal cost. But, a firm with market power may raise the price above the marginal cost by limiting its output. Market power arises in different contexts. One is monopoly, another is oligopoly and the other is monopolistic competition. Under monopoly, a single firm controls all matters related to production and selling. It is possible to raise the price above the marginal cost in the less extreme case of oligopoly. In monopolistic competitive market condition, though many firms produce a particular commodity, brands of a few companies are considered superior. For example, in India though a number of firms produce suiting materials, Raymonds, Gwalior and Vimal are regarded by many consumers as distinct and so their prices are not equal to the marginal cost. Failure to equate price with the marginal cost results in inefficient distribution of commodities, among consumers. The MRS of different consumers are not equal, the MRPT of different producers are not equal, the marginal rates of substitution between products and factors are not equal to their marginal rates of transformation and the price ratios of different products and factors are not equal to each other. Simply, if there is a market power, then Paretian optimum conditions cannot be achieved. A figure can help one to understand easily these ideas. (See figure 7.1). In the figure 7.1, all the conditions for maximum social welfare is achieved at the point E, but, points A and B are not optimal points. At point A, the good X is produced under monopoly condition while good Y is produced under perfectly competitive condition. But at point B, the good X is produced under perfect competition while the good Y is produced under monopoly. At point A, $MRPT_{xy}$ is less than $\frac{P_x}{P_y}$. At point B, $MRPT_{xy}$

is more than $\frac{P_x}{P_y}$. But consumer equates MRS_{xy} with $\frac{P_x}{P_y}$. Under 'A' situation, commodity 'X' is produced under monopoly, $MRPT_{xy} < MRS_{xy}$ and under 'B' situation, commodity 'Y' is produced under monopoly, $MRPT_{xy} > MRS_{xy}$. Thus, except point E, all other points indicate non-fulfilment of Pareto optimality.

7.6.2 Externalities

Another requirement for the validity of efficiency condition is that there are no external effects in consumption or in production. In other words, neither external economies nor diseconomies occur in consumption and in production. By externalities, we mean the presence of certain factors outside the pricing system, which are not captured by the market process. Externalities are market imperfections, where the market offers no price for service or disservice. The presence of externalities in consumption and production leads to Pareto non-optimality, mal-allocation of resources and consumption and production, to fall short of Pareto-optimality. In short, we can say externalities lead to divergence between social costs and social benefits and private costs and private benefits. The condition for Pareto-optimality is that there should not be any divergence between these costs and benefits.

7.6.2.1 Externalities in Consumption

It is assumed in perfect competition that utility is independent. In other words, for attaining Pareto-optimality it should be assumed that there is no externalities in consumption. The utility level of one consumer is independent of the consumption of the other consumer. But, in reality, it is not so. Consumption of one person affects the utility of others positively or negatively. In consumption, **bandwagon effect** and **Veblen effect** influence considerably the utility of others. Practically speaking, smoking of one person affects the utility of non-smokers. Using un-leaded petrol or maintaining clean environment around one's house increases the utility of neighbours. Consumption of a TV by a rich man increases the benefit of poor villagers if they are allowed to watch the programmes on that TV.

If there is no externalities in consumption, the marginal rate of substitutions will be equalized, $MRS^A_{xy} = MRS^B_{xy}$. If externalities are present in consumption, these rates cannot be equalized and so Pareto-optimality cannot be achieved. One can easily understand the above discussed points with the help of the following figure 7.8.

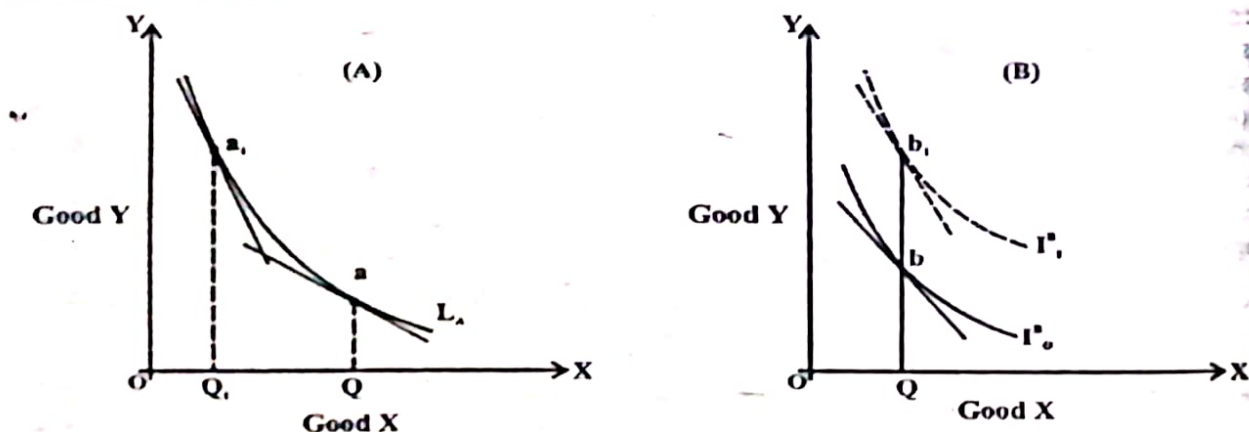


Figure 7.8

To get a clear understanding about external diseconomies in consumption it is necessary to have an assumption that consumer A, is not affected by the consumption of B, but A's consumption affects the utility of B as A is a smoker. If A increases his consumption of cigarettes 'X', B will be unhappy and if A decreases, B will be happy. We may take another example, if 'A' consumes more

water for his house use, it increases the mosquito problem as the waste water flows into an open ground, in which mosquitoes thrive. On the other hand, if he reduces the use of water, the pit will dry up and mosquito problem will be reduced and so B will be happy. Whatever may be the example, it is inferred that if the disutility creating consumption is reduced, then the social welfare will increase. In the above diagrams, the original positions are 'a' for A and 'b' for B. If 'A' reduces his consumption from OQ to OQ₁, then the utility of B will increase. Consumer 'B' is now in a higher indifference curve I₁^B, which is an upward shifted indifference curve. It means that he gets more satisfaction without any change in his budget or consumption. He consumes the same old quantity of OQ, in Panel (B). Though MRS_S at a₁ and b₁ are not equal, the social welfare increases in the situation because of the decrease in the consumption of good Y, by consumer A.

In the same way, in the case of external economies in consumption, an increase in consumption of 'A' increases the utility enjoyed by 'B'. For example, if A keeps a light burning in front of his house throughout the night it will benefit the opposite house owner who has no electricity connection and the others who use the path passing through (in front of) A's house. It can be explained by taking a₁ as the initial position of A and b as the initial position of B in figure 7.8.

7.6.2.2 Externalities in Production

As in consumption, there are two types of externality in production — external economies and external diseconomies. External economies occurs when productive activities of firms and industries benefit other firms and industries for which the contributing firms receive no payment. For example, when water is pumped out by a coal mine, the cost of pumping water by an adjacent mine is reduced. Or a new highway reduces the transport cost of individual firms. As these firms do not pay for this service, the marginal social cost is lower than the private marginal cost and private marginal benefit is less than the social marginal benefit. The private marginal cost is lower than the social marginal cost and private marginal benefit is higher than the social marginal benefit if there are external diseconomies in production. For example, the production of alcohol and its cost in terms of accidents and other inconvenience it creates. The private cost, which does not include the social cost like environmental damages, accidents due to drunken drivers and the inconvenience related to it, is less than the social cost, which takes all the social costs listed above. And so the social benefit is less than private benefit if there is external diseconomies in production. Here, the marginal social cost (MSC) is equal to the marginal private cost (MC) plus the marginal external cost (MEC).

Symbolically,

$$MSC_X = MC_X + MEC_X$$

The marginal social benefit (MSS) is equal to the marginal private benefit (MB_X) plus the marginal external benefit (MEB_X).

Symbolically,

$$MSB_X = MB_X + MEB_X$$

Now for our discussion, take the production of unleaded petrol for external economies and the production of alcohol for external diseconomies. The following figure 7.9 enlighten the readers further in this aspect.

the two firms must be equal. It can be easily adjusted and the total product will be completely exhausted if there is constant returns-to-scale. On the other hand, both decreasing returns and increasing returns create non-optimality problems.

In the case of decreasing returns, the total payment made is lower than the total income or total value of output produced. In other words, total product is not exhausted by factor payments. Thus, there is a distributional problem in a general equilibrium frame work. However, it is not a serious problem but requires only tedious mathematical calculations as we introduce non-linearity.

On the other hand, increasing returns to scale creates not only distributional problems but also equality problems because long-run marginal cost curve is below long-run average cost curve and product transformation curve is convex to the origin instead of concave to the origin. As we know, if LAC is above LMC, the firm incurs loss. The firm has to equalize LMC with MR and as AC is above MC the firm has to incur loss always. It finally leads to the collapse of the whole market system. Unless the loss incurred by firms is covered by government subsidies, such an output will not be produced. If subsidy is given, the government will have to seek for additional sources of revenue to pay the subsidy. But, in a competitive market it is almost impossible. This situation leads to disturbance to the Pareto-optimality and thereby, to the overall efficiency.

There will be no problem, if we have an unimportant increasing returns-to-scale. But, if the increasing returns to scale is strong, the production possibility curve will be convex to the origin as given in figure 7.10. However, two situations arise, viz.,

- (i) The curvature of the community indifference curve is greater than the curvature of the convex production possibility curve and
- (ii) The curvature of the PPC is greater than the curvature of the indifference curve,

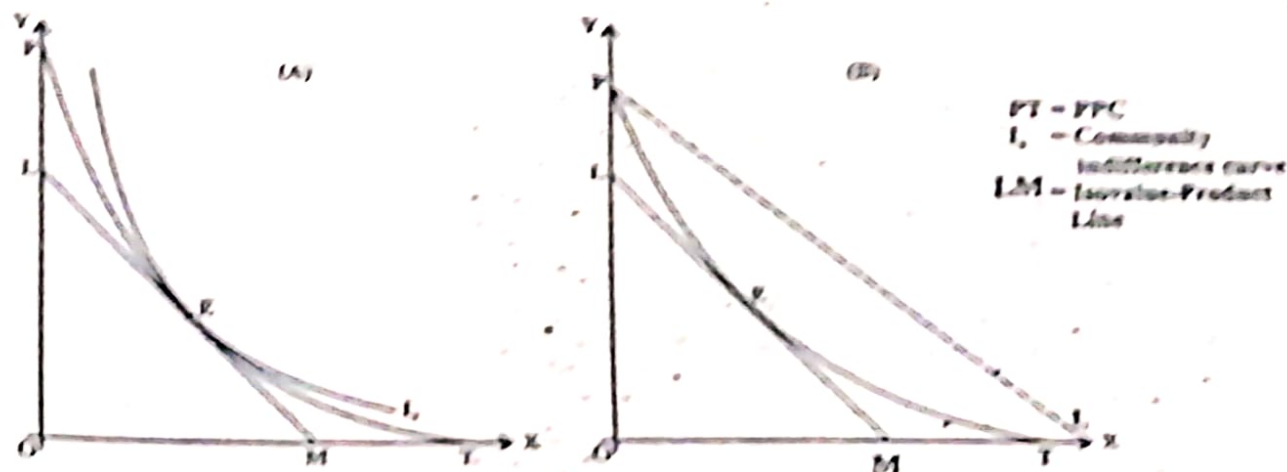


Figure 7.10

In panel (A) of figure 7.10, E is the equilibrium point where the isovalue-product line LM, the production possibility curve PT and the community indifference curve I_c are tangent. Here, the maximum social welfare is no longer associated with maximum value of output. At point E, the value

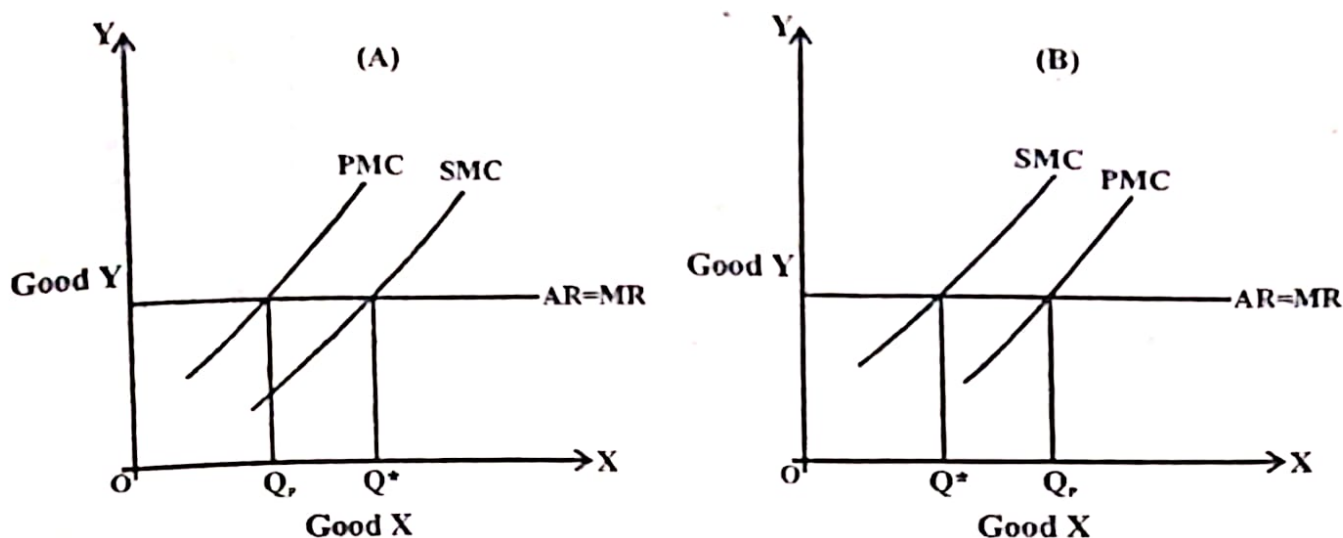


Figure 7.9

The two situations are explained in figure 7.9, panel (A) and (B). Panel (A) is meant for external economies and panel (B) for external diseconomies. The revenue curves are horizontal, $AR = MR$ as there is perfect competition in the market. In panel (A) by equalizing private marginal cost with private marginal benefit only OQ_P quantity is produced. But, if social benefits are included (SMS), then the ideal output is OQ^* . It means that less quantity is produced than the ideal if external economy prevails. In panel (B) by equalizing private marginal cost with private marginal benefit, OQ_P quantity is produced. But, the ideal output, based on social cost and social benefit should be only OQ^* . Thus, presence of external diseconomies creates a situation, where the actual quantity produced is more than the Paretian optimal output.

The whole economy can be explained with the help of a figure (see figure 7.1). In figure 7.1, the economy's ideal output (Pareto-optimal) is fixed at point E. But, for external economies, equilibrium will be reached at point A, where the quantity of commodity X produced is less than the ideal output. If there is external diseconomies, more of commodity X is produced and the equilibrium point is B. Hence, reallocation is necessary to satisfy the optimal condition.

For optimality, the marginal social cost must be equal to marginal social benefit. Symbolically, $MSC = MSB$. In a multi-product economy, equality must be maintained for all the commodities produced. Symbolically,

$$\frac{MSB_x}{MSC_x} = \frac{MSB_y}{MSC_y} = \dots = \frac{MSB_n}{MSC_n} = 1$$

But, in the real world because of the presence of externalities this condition may not be satisfied and so there is divergence between the ideal and the actual output.

7.6.2.3 Decreasing and Increasing Returns to scale

Ideal output is reached if constant returns to scale operates in the economy. But, if decreasing and increasing returns occur, then there will be some problems. Problems will be less for decreasing returns to scale than increasing returns to scale. Suppose, there are two firms producing a commodity by using only one factor, labour for reaching the ideal output point, the marginal product of labour in

the two firms must be equal. It can be easily adjusted and the total product will be completely exhausted if there is constant returns-to-scale. On the other hand, both decreasing returns and increasing returns create non-optimality problems.

In the case of decreasing returns, the total payment made is lower than the total income or total value of output produced. In other words, total product is not exhausted by factor payments. Thus, there is a distributional problem in a general equilibrium frame work. However, it is not a serious problem but requires only tedious mathematical calculations as we introduce non-linearity.

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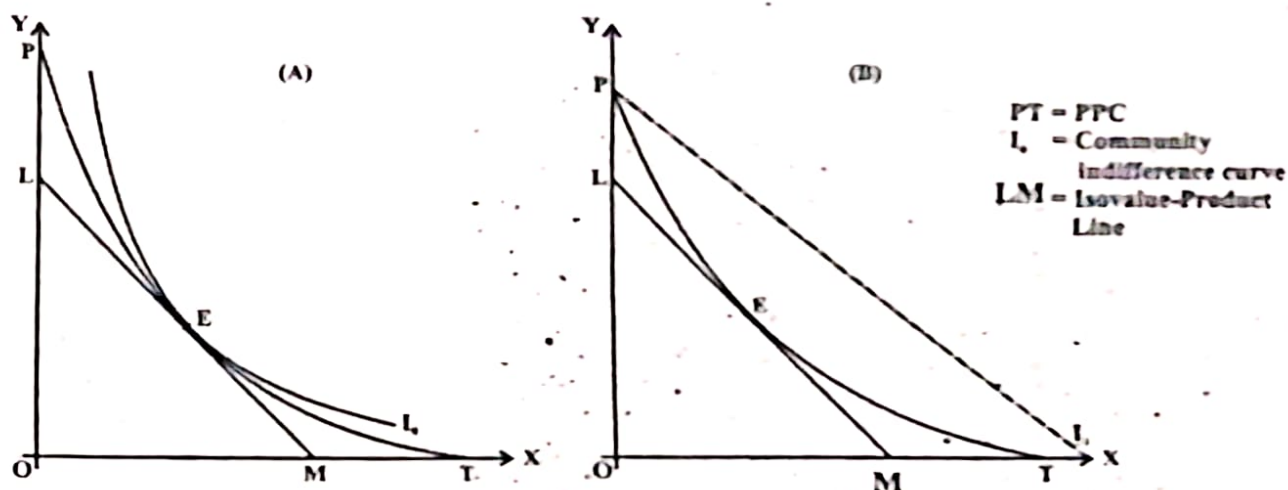


Figure 7.10

In panel (A) of figure 7.10, E is the equilibrium point where the iso-value-product line LM, the production possibility curve PT and the community indifference curve I_0 , are tangent. Here, the maximum social welfare is no longer associated with maximum value of output. At point E, the value

of output is at the minimum. Given $\frac{P_x}{P_y}$ the value of output will be maximized at either P or T.

Thus, in this situation, the social welfare is maximized but it is possible to get only a minimum value of output.

In panel (B) at the point of tangency E, both the social welfare and the value of output are minimum. Welfare maximization is attained only at point P, where the highest possible community indifference curve I_1 meets the PPC curve PT.

7.6.2.4 Uncertainty

Another reason for inability to obtain optimum welfare is uncertainty. Doubts arise about the efficiency of competition when there is uncertainty. What is meant by **uncertainty**? It is nothing but an economic state of affairs, where people don't know precisely about the consequences of their actions. In competitive market, we assume, that there is certainty and every one knows everything in the market as well as in the matters related to production activities. It means, that there is perfect knowledge about the market, both supply and demand and the consequences of every action undertaken. The next question that arises is, whether there is uncertainty and whether markets will break-down. The answer is simply no because the risk related to uncertainty can be reduced or eliminated. If there is risk, then there will be possibility for trading off the risk by, selling insurance to each other. The **first Welfare Theorem** tells that as long as a competitive insurance market exists, individuals bear risk in Pareto-efficient amounts. Another method to reduce risk is diversification. Any project's risk can be reduced by issuing shares and thereby diversifying the risk of one person to a large number of individuals. The stock market allows people to trade risk with one another, allowing a Pareto-efficient allocation of risk. Thus, the presence of uncertainty does not diminish the ability of competitive market in the efficient allocation of resources.

7.6.2.5 Non-existence and Incomplete Markets

The next assumption made by economists related to maximization of welfare is that there is market for every commodity and the market is complete. If there is perfect market, then it will be easier to fulfil all the three marginal conditions of Pareto. But, if a market for a commodity does not exist, we cannot expect the market to do the function of efficient allocation. In reality, markets for certain commodities are not in operation. The best example is market for insurance. Though insurance is a very important commodity, in the world of uncertainty, we cannot purchase insurance against certain events. It is not possible to purchase insurance against becoming poor. No insurance firm is **ready** to supply **poverty insurance**. If the insurance is supplied, the policy owner will decide not to work hard and hence the insurance company will have to monitor him. Monitoring is very difficult or impossible and so there is no market for poverty insurance. It means that if there is no market, then there will be no possibility for efficient allocation. On the basis this principle only it is inferred that maximum social welfare is not possible in the socialist economy where market does not do the allocation function.

Another reason for the failure to obtain maximum welfare is incomplete market or incomplete information - one party in a transaction has information that is not available to another. It is called **asymmetric information**. Asymmetric information is not good for the thriving of private markets. Lack of information or incomplete information certainly affects resource allocation. Unless one **knows** exactly all details about the product, he cannot maximize his welfare. The details about **asymmetric information** are discussed in chapter 9.

7.6.2.6 Public Goods

Another cause of market failure is the existence of public goods. Existence of public goods results in the non-attainment of Pareto-optimality, under perfect competition. A public good is one, whose consumption by one individual does not reduce its utility to any other individual. In other words, a good is considered to be a public good, if its use by an individual person or firm does not reduce its usage by other individuals or firms. For example, a breeze in the sea-shore. The traditional example is the light from the light house. Its use by one ship does not reduce the amount of light available to other ships. Simply, consumption of a public good is non-rival in nature. Another character of public goods is that they are not subject to the exclusion principle or non-excludability. Exclusion principle refers to a system where those who have not paid are excluded from the enjoyment of the service. But, public goods are not subject to the exclusion principle, which means that their benefits are available to all, irrespective of whether they pay for them or not. In other words, in the case of public goods, the exclusion of non-payers from the benefits is impossible and in some cases extremely costly. For example, collecting money from the ship owners, who make use of the light from light house, is impossible but money can be collected from the persons who use the bridge by establishing a toll gate.

If individuals can enjoy the benefit of a public good, without paying for it, they have every incentive to ride free. A free rider is one who derives or gets benefits of a commodity or a service without paying for it. When everyone tries for a free ride it will result in imbalance and market failure. Even if it is possible to exclude non-payers, it is not possible to reach the Pareto-optimality point.

7.6.2.7 Indivisibilities

The Pareto optimality is based on the assumption of complete divisibility of products and factors. In reality, they are not perfectly or infinitely divisible. In the production world, there are small firms and large firms. The economy has attained the equality of the MRPT and the MRS of the two commodities among the consumers. In this situation, the three marginal conditions of Pareto optimality are fulfilled. But, the use of resources is inefficient and social welfare is not maximized due to the indivisibility of production methods and factors of production. The small firms cannot take advantage of the lower cost of the large-scale production techniques. Large firms produce a greater amount of output with the same amount of resources and small firms produce less output with the same amount of resources because of inapplicability of the large-scale production techniques.

Thus, Pareto optimality is not possible to attain when –

- (i) there is market power
- (ii) there are externalities in production and in consumption
- (iii) there are strong economies of scale which result in convexity in the PPC
- (iv) there is uncertainty
- (v) there is no market or incomplete market
- (vi) there are public goods and
- (vii) there are indivisibilities in the production function.

7.7 Theory of Second Best

The theory of Second Best was propounded by R.G. Lipsey and Kelvin Lancaster (1956). So it is also called Lipsey Lancaster theorem. The theory says that if a constraint operates in the economy in the path of fulfilling the Pareto optimal conditions and the constraint cannot be overcome, then the best policy for social welfare maximization is to deviate from the optimal conditions, without fulfilling the remaining conditions. If all the Paretian optimality conditions are satisfied, then it will be the **first best solution**. But, in reality it is not possible as one or more of the Paretian marginal conditions cannot be satisfied due to some political, social or economic constraints. If that is the real situation, then maximization of social welfare is unattainable. As the first best solution is not possible, it is better to search for the **second best solution**. This is the gist of the theory of Second Best.

The economists have the view that when some constraints are operating in the economy, it is not necessary that the satisfaction of the remaining Pareto marginal conditions, improves social welfare. In other words, the theory states that if one or more of the first order conditions cannot be satisfied because of the presence of some constraints, it is neither necessary nor desirable to satisfy the remaining conditions. Simply, social welfare can be maximized by deviating from Pareto optimality conditions. By looking into the figure 7.11, one can easily understand the first best solution and the second best solution. Even if it is not possible to reach the ideal output or ideal consumption, it will be possible to know where to reach, to maximize the social welfare.

In the absence of any constraint, the first best solution in the figure 7.11 is E^* , where the society's welfare is maximized. The highest possible indifference curve is I_2 . A constraint operates in the form of income line or budget line LM . In this situation, it is not possible to reach the maximum indifference curve I_2 . Hence, it is better to move inside the PPC curve PT . The Paretian optimality conditions are, not, satisfied at the point a , and the social welfare attained is shown by the community indifference curve I_0 . But, the social welfare at point E_1 is more than at the point a , because the welfare attained at point E_1 is represented by I_1 , which lies above I_0 . Simply, the welfare level at point E_1 where Pareto optimality conditions are not satisfied is more than at point a , where other but not all, Pareto optimality conditions are satisfied. Hence, it is true that it is not necessary to try to fulfil other marginal conditions, if there are some constraints in achieving the first best solution.

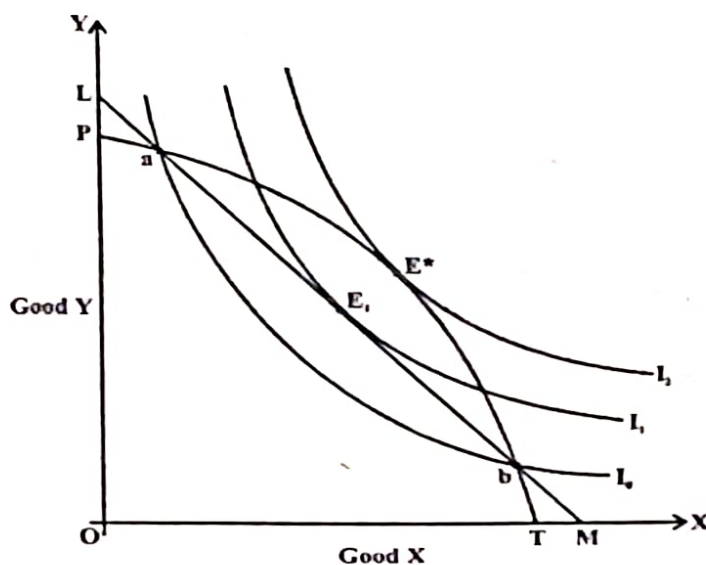


Figure 7.11

Now, take an example to explain this situation. In the economy, two goods 'X' and 'Y' are produced. Commodity X is produced under monopoly conditions and the market for commodity Y is perfectly competitive. Here, monopoly acts as a constraint in the way of attaining Paretian equilibrium. In such a situation, it is good to the economy, maximizing social welfare to deviate more and more from perfect competition, in the rest of the economy. An attempt to ensure the fulfilment of

Pareto optimal conditions causes a fall in social welfare. It is based on the argument that if one commodity is produced by monopolist and another by a perfectly competitive producer then there is deviation between MRS and MRTS ($MRS > MRTS$). Hence, an efficient output is not produced and allocation of resources is not optimal. But, if both the commodities are produced under monopoly condition, then there will be a possibility for reaching the equality condition, i.e., $MRS = MRTS$.

7.8 Arrow's Impossibility Theorem

The social welfare function developed by Bergson is a method for finding out the social welfare from individuals' welfare. The function tries to arrive at a social ordering of feasible social states from individuals' ranking of these social states in terms of their desirability. The social ranking of alternative economic situations can be made only by making interpersonal comparisons of utility. Kenneth J. Arrow in his book 'Social Choice and Individual Values' (1951) states that even if individual preference ranking or ordering, satisfies the property of consistency, it may not give us a consistent ranking for social preference. A consistent social ranking of economic situations may not exist, even if one uses the democratic principles of majority voting. Simply saying, there is impossibility of obtaining the social welfare function even if individual preferences are consistent. He suggests seven minimum conditions or criteria, which social choices must satisfy in order to reflect preferences of individuals.

- (i) **Completeness:** The social ranking must rank all feasible economic situations.
- (ii) **Consistent:** The social ranking must be consistent. Consistency refers to the requirement that individuals have preferences that are fully defined, i.e., every alternative is ranked in relation to every other.
- (iii) **Transitivity:** The ranking must be transitive. In other words, social choices must satisfy the condition of transitivity. If a society prefers X to Y and Y to Z, then the society must prefer X to Z. The social preferences are completely ordered.
- (iv) **Responsiveness:** Social choices must be directly related to individual preference, i.e., social choices must change in the same direction as individual choices. In other words, the social ranking must be based on individual preferences and must respond to individual's preferences.
- (v) **Non-dictation:** The social ranking must not be imposed or dictated. It should not be imposed by one individual within the society as it depends on individual preferences.
- (vi) **Non-imposition:** The social ranking must not be imposed by customs or forces from outside the society but derived from individual preferences.
- (vii) **Independence of Irrelevant Alternatives:** If any one alternative is excluded, it will not affect the ranking of the other alternatives.

A social welfare function exists only when the above mentioned axioms are fulfilled simultaneously. Arrow demonstrates that it is not possible to satisfy all these conditions but there is good chance for getting at least one condition violated. It is very difficult to obtain a transitive social choice for each set of individual preferences, without violating at least one condition. It is said that social choice is inconsistent or undemocratic. This view has come to be known as the Arrow's Impossibility Theorem.

The following example shows that such a all conditions-fulfilled social welfare function may not exist. Suppose, an economy consists of 3 individuals A, B and C and there are three alternative states of affairs X, Y and Z. The individuals' rankings are as follows.

A $\rightarrow$ X > Y > Z

B $\rightarrow$ Y > Z > X

C $\rightarrow$ Z > X > Y

It can now be numerically denoted by assigning 3 for their first choice, 2 for second choice and 1 for third rank.

Table 7.1: Individual Preference

Individuals	Alternative		
	X	Y	Z
A	3	2	1
B	1	3	2
C	2	1	3

The table 7.1 and the figure 7.12 show that A prefers X to Y to Z and B prefers Y to Z to X and C prefers Z to X to Y and each individual has consistent preference. But, the majority voting in a democratic state leads to intransitive social patterns. A and C prefer X, to Y, A and B prefer Y to Z and B and C prefer Z to X. All the three alternatives X, Y & Z are preferred by two of the three individuals. This inference is known from the multi-peaked pattern of the curves indicating individuals' preference. It is very clear, that a democratic voting system has not given any equilibrium. It can easily be inferred that the democratic process of ranking leads to a contradictory welfare situation. This is called **voting paradox**. The voting paradox arrives whenever individual preference are not single peaked.

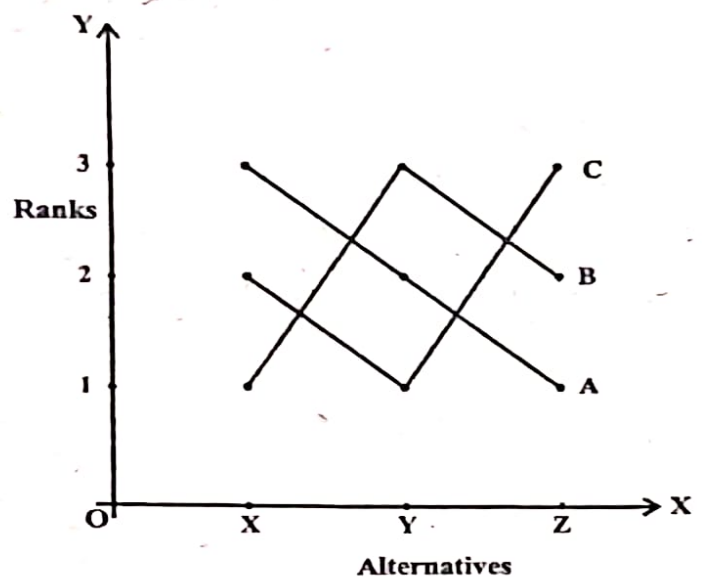


Figure 7.12

7.8.1 Criticisms

Arrow's impossibility theory is also not free from limitations. The important limitations are as follows:

Arrow talks only about democratic voting, but fails to talk about a system where there is permission for buying and selling of votes as well as for taking a unanimous decision.

- Arrow's way of explaining the model is also very seriously criticized. It is nothing but a theorem of mathematical politics.
- Another limitation of Arrow's theorem is that it is only a theorem of decision-making rather than a theorem of social welfare maximization.
- Arrow's theorem is also not free from value judgements or inter-personal comparisons.

7.9 Rawl's Theory of Justice

John Rawl, known mainly for his theory of justice, is one of the most important political philosophers of the second half of the 20th century. His theory of Justice as fairness or simply theory of justice, develops principles of justice to govern a modern social order. He has put forward his ideas on social justice first in the book 'A Theory of Justice' (1971), then in his second book 'Political Liberation' (1996) and at last in the book 'Justice as Fairness: A Restatement' (2001). The theory of justice provides a framework that explains the significance of political and personal liberties, of equal opportunity and cooperative arrangements that benefit more the less advantaged persons of society. Darrel Moellendorf considers Rawl's conception of justice as an associational conception because Rawl's initial concern with justice is related to relationship between persons within an association (society). There is a system of cooperation over time, from one generation to another. The members of the society enter into the relation when they are born and exit from it only at death.

Rawl developed a conception of justice from the perspective that persons are free and equal. Their freedom consists in their possession of two moral powers namely — (i) a capacity for a sense of justice and (ii) a capacity for a conception of good. A sense of justice refers to the individual's capacity to understand, to apply and to act from the public conception of justice. This sense indicates the willingness of individuals to act in relation to others. A conception of good is a conception of what is valuable in human life. Persons have the above mentioned two moral powers and so they are responsible for their actions. This is called **political conception**.

A political conception of justice involves three ideas, the original position, the veil of ignorance and the derivation of two principles of justice. Rawl, further, assumes that all citizens are reasonable in their behaviour. They are prepared to offer one another fair terms of social cooperation. They also agree to act on these terms, sometimes at the cost of their own interest, provided that others also accept these terms. They do this freely and are not dominated or manipulated by or under the pressure of an inferior political or social position. Rawl calls this **the criterion of reciprocity**. They also recognize and are willing to bear the consequences of the burden of judgement. They also affirm the reasonable comprehensive doctrines. Rawl notes three features of the doctrine. They are, (i) it covers the major religious, philosophical and moral aspects of human life, an exercise of theoretical reason, (ii) it determines the values, to regard as highly significant and how to weigh them against each other when they conflict, an exercise of practical reason and (iii) it evolves over time in the light of what it sees as good and sufficient reason. Reasonable persons allow others freedom to explore religious and spiritual options even if these options differ from their own.

7.9.1 Theory of Justice

All criteria of Pareto, Kaldor and Hicks and Scitovsky, analyse the concept of welfare in terms of efficiency effect, by completely neglecting equity or distributive justice. The separation of efficiency and equity considerations in decision-making presumes that there is always a conflict between the objective of efficiency and the goal of equity. But, in many situations, by improving efficiency in the use of resources, equity can also be improved and an improvement in equity also